Fabricated and inaccurate citations generated by large language models for biomedical topics: a systematic review and meta-analysis

Review 2 of the Cochrane-style clinical AI meta-analysis series.

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Protocol version 1.0, 22 October 2026 (protocol.md, this series). Registration: no PROSPERO registration number was issued before analysis. A PROSPERO-ready form was prepared before screening (prospero-form.md) and an OSF record is the transparent parallel route. No registration is backdated or implied anywhere in this review.

Version note: this is the corrected version of the review first published on 22 October 2026. A robustness pass on 23 October 2026, after first publication, closed three process gaps that the first version disclosed (a coded adjudication rule at screening, a single extraction with verification, and a mechanical risk of bias adjudication rule) and corrected the dataset and every pooled estimate. The amendment is logged in the Methods (section 1) and in llm-team-log.md. The first version stands as published; nothing in it has been silently altered, and the first version's analysis files are preserved in the repository.

Methods standard: Cochrane Handbook for Systematic Reviews of Interventions, version 6.5 (2024), followed in spirit and structure. Reporting follows PRISMA 2020 and PRISMA-S. This is a Cochrane level review. It is not a Cochrane review and was not submitted to Cochrane, because Cochrane requires at least two authors and this is a single author review with LLM assistance under human accountability, reported in full in the declarations and the team log.

Scope note: this review covers evidence synthesis and research integrity only. It makes no clinical recommendation and no medication recommendation, and it gives no advice for the care of any individual patient.

Abstract

Rationale. Large language models produce reference lists that look sound. A share of the references do not exist, and a further share contain errors that stop a reader finding the source. Primary studies have reported these rates study by study, with different models, topics and definitions. No published synthesis pools them on their shared denominator, references generated.

Objectives. To estimate, from generation experiments, the proportion of LLM generated biomedical references that are completely fabricated, the proportion with any major inaccuracy, and the proportion fully accurate, and to examine how those proportions vary by model family, retrieval use, publication type and study level pooling.

Search methods. PubMed, Europe PMC, OpenAIRE, Crossref and an OpenAlex test page were searched on 22 October 2026, with exact strings, dates and hit counts in the search log. arXiv and Semantic Scholar were attempted and blocked by rate limits and are logged as blocked. Embase, Scopus and Web of Science strings were prepared and are marked run pending access. No language limit was applied at search. In the robustness pass (23 October 2026), full OpenAlex retrieval and further arXiv and Semantic Scholar retries were attempted and blocked again after retries (HTTP 429), and systematic citation chasing of all included studies and two competing reviews added 489 new records, of which 2 studies were included.

Eligibility criteria. Primary studies that prompted one or more LLMs to generate references on a biomedical or clinical topic, reported the number generated, and verified each reference against a named source. Large scale audits of the published literature were eligible as a separate stratum and were never pooled with generation experiments. Editorials, reviews with no original experiment, detection tools, citation screening studies and reports with no denominator were excluded.

Outcomes. Critical: proportion completely fabricated; proportion with any major inaccuracy including fabricated; proportion fully accurate on all checked fields. Important: minor errors by field, non-resolving or wrong DOIs, error by list position where reported.

Risk of bias. An adapted tool specified before extraction (rob-forms.md), with eight domains covering topic and prompt sampling, model and prompt reporting, generation

settings, verification source, verifier independence, duplicate or blinded verification, partial match handling and selective reporting.

Synthesis methods. Random effects meta-analysis of proportions in Python (numpy, scipy, pandas), logit transformation, REML tau squared, prediction intervals where 10 or more arms pooled. Model arms within studies were kept visible; a per study pool was run as a sensitivity check for every outcome.

Included studies. 16,474 records were screened (15,985 from the database search and 489 from citation chasing). Sixty three generation experiments (148 model arms) and one published literature audit met the rules.

Results of syntheses. Completely fabricated: pooled 19.4 percent (95% CI 14.9 to 24.8), 113 arms, 50 studies, 22,461 references, I squared 97.1 percent, prediction interval 1.0 to 85.6 percent, GRADE very low. Fully accurate: pooled 37.0 percent (27.2 to 47.9), 58 arms, 28 studies, 14,224 references, I squared 98.2 percent, GRADE very low. Any major inaccuracy: pooled 16.5 percent (5.7 to 39.2), 20 arms, 14 studies, 2,668 references, I squared 96.7 percent, GRADE very low. Per study sensitivity pools gave 24.7 percent fabricated and 38.2 percent fully accurate. Arms with retrieval, browsing or database access active pooled 5.8 percent fabricated (26 arms) against 31.0 percent for arms with retrieval off (50 arms). The audit stratum, reported apart, found fabricated citations in published papers rising from about 1 in 2,828 papers in 2023 to about 1 in 277 in early 2026.

Authors' conclusions. In these experiments, roughly one generated biomedical reference in five did not exist, and just over one in three was fully accurate. The spread between models and setups was wider than any average, so the pooled figures describe a literature, not a property of a named product. Verification of every generated reference against a database is the only step in this literature that consistently separated usable lists from unusable ones. Certainty is very low throughout, mainly because verification conduct was poorly reported and heterogeneity stayed high after every pre-specified subgroup.

Funding. None. **Registration.** None issued before analysis; see the title page note.

Plain language summary

The plain language summary is filed as plain-language-summary.md and reproduced in the repository README. In short: of every 100 references a model generated in these experiments, about 19 did not exist and about 37 were fully accurate. Rates varied sharply by model and by whether web access was on. A separate audit found fabricated citations

already sitting in published papers, and rising fast between 2023 and 2026. The quality of the underlying studies was weak enough that all main results carry very low certainty.

Summary of findings

GRADE, adapted for proportion and prevalence evidence as set out in the protocol, applied per critical outcome. Downgrade reasons were stated in the protocol before analysis.

Outcome	Arms	Studies	References checked	Pooled proportion (95% CI)	I squared	Certainty	Reasons
Completely fabricated	113	50	22,461	19.4% (14.9 to 24.8)	97.1%	Very low	Risk of bias down 1 (30 of 63 studies at high risk on the adapted tool, verification conduct often unreported); inconsistency down 1 (prediction interval 1.0 to 85.6%); indirectness down 1 (experimental lists

							stand in for real manuscript drafting)
Any major inaccuracy, including fabricated	20	14	2,668	16.5% (5.7 to 39.2)	96.7%	Very low	Risk of bias down 1; inconsistency down 1; imprecision down 1 (wide interval)
Fully accurate on all checked fields	58	28	14,224	37.0% (27.2 to 47.9)	98.2%	Very low	Risk of bias down 1; inconsistency down 1 (prediction interval 1.9 to 94.8%); indirectness down 1

Outcomes with thinner data: minor errors by field and bad DOI counts were extracted where printed and are reported in the Results without pooling, because field level denominators differed between studies. Error by list position was reported by too few studies to pool and is described narratively.

Background

Description of the problem

A reference list does quiet work. It tells a reader that a claim rests on a paper that exists, that the paper says what the writer says it says, and that the reader can go and check. Large language models write reference lists that keep the shape of that promise and break its

content. The authors are plausible because the model has seen thousands of real author names in the field. The journal is real. The year fits the topic. The paper does not exist, or it exists with a different title, or the DOI resolves to another article entirely.

This is not a rare glitch at the edge of the technology. When Chelli and colleagues asked GPT-3.5, GPT-4 and Bard for references for systematic reviews in 2024, the hallucinated shares were 39.6 percent, 28.6 percent and 91.4 percent. When researchers in 2026 asked six models for radiology references, 480 per model, fabrication ran from 3 percent for one model to 61 percent for another in the same study, on the same topics, with the same verification (Gunes 2026, extracted in this review). A tool that drafts a manuscript section and appends a bibliography is now a standard feature of the writing software researchers open every day. The bibliography is the part nobody retypes by hand, which is exactly why errors in it travel.

The published literature already carries the marks. Yin and colleagues audited about 111 million references across 2.5 million papers on arXiv, bioRxiv, SSRN and PubMed Central and estimated about 146,932 non-existent references in 2025 alone, with a sharp rise after mid 2024. Topaz and colleagues, in a 2026 Lancet letter, audited 2.5 million biomedical papers and found fabricated references rising from about 1 in 2,828 papers in 2023 to about 1 in 277 papers in the first weeks of 2026. Those audits measure what reached print. They cannot say which step produced the errors, because a published reference list mixes human error, copy editing error and model generation. The generation experiments can answer the narrower question, and until now nobody had pooled them.

Description of the intervention

There is no intervention in the clinical sense. The unit under study is a generation setup: a model, at a version, with stated or unstated settings, with or without web access or retrieval, given a prompt or a task that asks for references on a biomedical topic. The comparator, where one exists, is another setup in the same study run on the same topics. Most primary studies are bench experiments, not trials, and this review treats them that way.

Why it is important to do this review

Editors, reviewers and readers currently argue from single studies. One paper finds 7 percent fabrication from a current model on guideline topics (Bernstein 2026, 195 of 2,736) and the next finds 66 percent from another model in another specialty (UcarGundogar 2026, DeepSeek R1 arm, 98 of 149). Both can be accurate reports of their own setups. A pooled estimate with its heterogeneity stated, split by model family and by retrieval, gives the argument a shared set of numbers. It also gives the methods literature a target: if

verification conduct is the weak point, the pooled result should show where the weakness sits. Review 1 of this series covered ambient AI scribes. The gap ranking behind the series placed LLM hallucination second, on the condition that a review take one task with one denominator. Citation generation is that task.

Objectives

To assess, from primary generation experiments, the proportions of LLM generated biomedical references that are completely fabricated, that carry any major inaccuracy, and that are fully accurate, and how those proportions vary by model family and generation, retrieval use, specialty and topic type, list length, publication type and model year.

Research questions: RQ1, what proportion of generated references are completely fabricated? RQ2, what proportion carry any major inaccuracy including fabrication? RQ3, what proportion are fully accurate on all checked fields? RQ4, how do these proportions differ by model family and generation? RQ5, how do they differ with retrieval, browsing or a reference database active against none? RQ6, how do they differ by specialty, established against emerging topic, and number of references requested? RQ7, what happens to generated errors after publication, as measured by the separate audit stratum?

Methods

1. Differences between protocol and review, and the post-publication robustness amendment

Amendment, 23 October 2026 (after first publication; this entry is dated and is not backdated). The first version of this review (22 October 2026) disclosed three step-downs from the full Cochrane-level process in its own text: screening disagreements were settled by a coded adjudication rule rather than by reading; extraction was a single extraction with a verification pass rather than two independent extractions; and risk of bias disagreements were settled by taking the more cautious judgement rather than by reading. The author directed that all three be closed, and a robustness pass was run on 23 October 2026. In that pass: (1) an adjudicator read all 913 stage 1 disagreement records and passed 190 to full text, of which 161 had never been sought; 95 were retrieved and assessed, and 37 further generation studies were included; (2) the search was expanded by systematic citation chasing (979 candidates from the reference lists and citing works of the included studies and of two competing reviews, run through Europe PMC and Crossref; 489 genuinely new records, dual screened with adjudication by reading; 10 reports sought, 7 retrieved, 2

studies included), and full OpenAlex retrieval plus renewed arXiv and Semantic Scholar attempts were made and blocked again after retries (HTTP 429), logged as blocked and never as searched; (3) every included study received a second, fully independent extraction, the two passes were compared field by field by code (825 field cells compared on matched arms of the original studies where both passes had an entry; most string level differences were wording in free text fields such as definitions notes and source locations; on the numeric inputs to the pools, 25 of 201 compared cells differed), and every discrepancy was settled by re-reading the source, with the settled value, the source location and the wrong pass recorded in robustness-discrepancy-log.md; (4) every risk of bias disagreement, in the original set and the new studies, was settled by an adjudicator reading the paper, and the more-cautious-wins rule was not used. The pass also found, on re-reading, that one included study (Jedrzejczak 2024) is the preprint of another (Pastucha 2026, whose acknowledgements name the preprint as an earlier version of the same manuscript, with identical data for the shared arms); the two reports are now merged at study level and counted once, under Pastucha 2026. Pooled estimates were re-run on the corrected dataset; the first version's figures are preserved in the repository (robustness/v1/) and the before and after estimates are reported side by side in the team log. This corrected version replaces the first version's numbers everywhere; the first version itself stands as published and has not been edited.

Four differences from the protocol, each logged with its reason. First, the strict second pass over abstracts was applied at the full text stage rather than before retrieval, because open access retrieval for all 122 stage 1 survivors was automated (Europe PMC then Unpaywall) and cost less than a separate pass; every retrieved report still received a full text decision with one exclusion reason. Second, analysis ran in Python rather than R, because no R installation was available in the working environment; the protocol named Python with a logit model as the stated fallback, and the code is published with the review. Third, the model year subgroup could not be run as specified: primary studies rarely printed a model year apart from the model version, so that subgroup is reported as not estimable rather than rebuilt from publication year, which would answer a different question. Fourth, list length subgrouping used references requested where a study printed it; arms without a printed requested count sit in a not reported group and are not assigned a list length from their generated count. No outcome definition changed after extraction began. No eligibility rule changed.

2. Criteria for considering studies for this review

Types of studies: primary generation experiments of any design, and preprints of the same, with original data. Published literature audits formed a separate stratum under their own

extraction section. Systematic reviews, editorials and commentaries were used for background and reference chasing only.

Types of participants: not applicable in the clinical sense. The sampled units are topics, prompts and reference lists in biomedical and clinical subject areas. Studies in other fields were included only where a biomedical subset could be extracted apart (Mugaanyi 2024, natural sciences subset). Cross disciplinary studies that reported only discipline level aggregates with no biomedical split were excluded (for example the Walters and Wilder experiment, captured in this search, reports by work type rather than topic).

Types of generation setups: any LLM or LLM based assistant asked to produce references or citations, with model, version, settings and retrieval status extracted where printed.

Outcomes and definitions: a reference was completely fabricated where no matching publication existed under the study's verification. A major inaccuracy was an error that prevents retrieval or misattributes the source, including fabricated references, a wrong DOI that resolves to a different paper, or a title or author error the study graded as major. A reference was fully accurate where every field the study checked was correct. Definitions were extracted as printed for every study; a study entered a pool only where its definition mapped exactly to the review definition, or where its printed components summed to the review definition, with the mapping logged per row in extraction.csv.

3. Outcome measures

Critical outcomes are the three proportions in the summary of findings. Important outcomes: minor error rate; minor errors by field (authors, title, journal, year, volume or pages, DOI); proportion with a DOI that does not resolve or resolves to a different paper (for example Athaluri 2023, 69 invalid DOIs among 178 generated references; Adiguzel 2026, DOI errors in 94 of 150 ChatGPT 5 references against 14 of 150 for Claude Sonnet 4.6); error by list position where a study printed it.

4. Search methods for identification of studies

Databases and sources searched on 22 October 2026: PubMed through NCBI E-utilities (372 retrieved), Europe PMC REST API with cursor paging (14,090 retrieved, string not restricted to title and abstract, a breadth decision that is logged and that screening absorbed), OpenAIRE (10), Crossref bibliographic and posted content queries (1,000 records retrieved from each; the total-results figures of 1,507,235 and 122,445 are relevance match counts and were never treated as result sets), and an OpenAlex test page (25 records; full retrieval not run in the first pass). arXiv timed out and then returned HTTP 429; Semantic Scholar returned HTTP 429 on its test call. Both are logged as attempted and blocked. Embase,

Scopus and Web of Science were not searched: no licensed access was available, and ready to run strings for each sit in search-appendix-pending.md, marked string prepared, run pending access. This review does not claim those databases were searched. In the robustness pass (23 October 2026), full OpenAlex retrieval was attempted three times and blocked (HTTP 429; the shared network budget was exhausted), arXiv was retried by two routes (timeout, then HTTP 429) and Semantic Scholar by two routes (HTTP 429 on both); all are logged as blocked after retries, never as searched. Systematic citation chasing was then run on the reference lists and citing works of all 26 first version included studies and of the two competing reviews named in the Background (Basu and Huynh 2026; Topaz 2026), through Europe PMC with a Crossref fallback: 979 raw candidates, 454 already in the search set, 36 within-set duplicates, and 489 genuinely new records, which were abstract-enriched from Europe PMC (404 of 489) and dual screened; 2 further studies entered through chasing. Full strings as run sit in search-log.md, with the robustness pass in its own dated section.

5. Data collection and analysis

Selection of studies. Two LLM screeners with different written rules (A strict, B sensitive) screened all 15,985 deduplicated titles and abstracts apart. Agreement was 94.29 percent; Cohen kappa was 0.13, a prevalence effect described in the team log. In the first version, a coded adjudication rule passed 122 records to full text. In the robustness pass, an adjudicator read every one of the 913 disagreement records (all of them B-only) and passed 190 to full text (28 include, 162 maybe); agreement between the real adjudication and the coded rule, on a pass or fail basis, was 80.0 percent: the coded rule had passed 22 records the adjudicator excluded, and had dropped 161 records the adjudicator passed. None of the 22 had reached the first version's included set, and none of the 25 first version studies was excluded by the adjudicator, who called all 12 of their disagreement-set records include or maybe. Full texts were sought for all 293 reports across the three streams (122 first pass, 161 re-adjudicated, 10 from chasing); 178 were retrieved open access and 115 were not retrieved. Full text decisions were made by reviewers reading each text, each exclusion carrying one reason.

Data collection and management. Extraction followed extraction-form.md: 57 fields across identity, topic, per arm model and setup, verification, definitions as printed, counts as printed, funding and conflicts, plus a separate audit section; the analysis dataset carries the 25 fields that feed the pools. Percentages were never back calculated into counts unless the count was also printed, with one stated exception: where a paper printed percentages against a stated denominator and exactly one whole count reproduces the printed percentage (or, for Seifi 2026, integer percentages against a printed denominator of 100 per

model whose derived counts summed exactly to the paper's printed pooled totals), the derived counts were entered and flagged derived in every affected row. In the robustness pass, a second extractor team re-extracted every included study from its full text without sight of the first extraction; the passes were compared field by field by code and every discrepancy was settled by a reader re-reading the source, with the outcome logged in robustness-discrepancy-log.md. Of the original studies' pooled inputs, settlements blanked 11 cells the first pass had filled on definitions that did not map exactly (including the fully accurate counts of Civelekler 2026 and the major inaccuracy counts of Athaluri 2023 and Dhaimade 2026), filled 5 cells the first pass had left blank (including Metze 2024, whose percentages at denominator 260 derive uniquely to counts), and removed one duplicate study (Jedrzejczak 2024, merged into Pastucha 2026).

Assessment of risk of bias. The adapted tool in rob-forms.md, specified before extraction, with eight domains. Two assessors judged every generation study apart. In the first version, disagreements (56 domain and 5 overall, of 200 domain judgements) were adjudicated by taking the more cautious judgement; in the robustness pass an adjudicator re-judged every disagreement by reading the paper and both rationales, and the same process was run for the new studies. Two of the 25 original studies changed overall category (Athaluri 2023 and Sawamura 2024, high to some concerns). A reader can rebuild the assessments from the assessor sheets and rob-assessments-v2.csv. The tool is adapted; it is not RoB 2 and not QUADAS-2.

Measures of effect. Proportions per model arm, with the arm as the primary unit and the study as a clustering level handled by a per study sensitivity pool. No odds ratios or comparative effect measures apply, because most primary studies have no non-LLM comparator.

Synthesis methods. Random effects meta-analysis of proportions, logit transformation, REML estimate of tau squared, in Python (analysis.py; numpy, scipy, pandas). Prediction intervals are reported where 10 or more arms pool. Reporting bias tests were not run: funnel based tests behave badly for single proportions and the protocol limited them to pools of 10 or more units with care; given I squared above 95 percent in every main pool, a funnel plot would describe heterogeneity, not reporting bias, and none is presented as evidence either way.

Subgroup and sensitivity analyses. Pre-specified subgroups only: model family; retrieval, browsing or RAG against none, with not reported kept apart; specialty and topic type in narrative form where arms were too few to pool; list length by references requested, 10 or fewer against more than 10, where printed; peer reviewed against preprint; model year not

estimable (see differences, above). Sensitivity analyses: per study pooling; peer reviewed reports only (preprints removed); leave one model family out; studies at low or some concerns risk of bias only. A strict against lenient fully accurate sensitivity could not be run because no included study printed both definitions for the same arms; that gap is reported rather than filled.

Certainty of the evidence. GRADE adapted for proportion evidence (protocol.md), per critical outcome, with downgrade reasons fixed in advance: risk of bias in verification, inconsistency where model and topic mix stays unexplained, indirectness where experimental lists stand in for real drafting, and imprecision.

6. Equity considerations

Two equity points arise. First, verification labour falls unevenly: a researcher with library support and database access can check a generated list in minutes per reference, while a researcher without that support carries the full cost, and fabricated references in print then mislead readers in settings with the least access to check them. Second, several included experiments found worse accuracy on emerging or low resource topics, the topics where local researchers most need drafting help. No included study reported results by author country or language of the prompt in a form that could be pooled, so both points are stated as context, not as findings of this review.

7. Consumer involvement

No patients or members of the public were involved. The stakeholders for this question are authors, editors, reviewers and librarians; the conference route for this review targets a library audience for that reason.

Results

Description of studies

Records identified: 16,497 retrieved across all database runs, plus 979 citation chasing candidates of which 489 were new records. Duplicates removed: 512 from the database set and 36 within the chasing set. Records screened: 16,474. Records excluded at title and abstract after dual screening and adjudication by reading: 16,181. Reports sought: 293. Reports not retrieved: 115. Reports assessed at full text: 178. Reports excluded at full text: 111, with one reason each; the largest groups were reviews and editorials with no original experiment, duplicate reports of an experiment already captured, retrieval or detection tools with no generation experiment, non-biomedical reports with no separable biomedical data, and reports whose retrieved file did not correspond to the record and whose correct

text could not be obtained. Reports included: 66 generation experiment reports covering 63 unique studies (one first version report, Jedrzejczak 2024, proved to be the preprint of Pastucha 2026 and is counted once, under Pastucha 2026), and 1 audit report. The full flow sits in prisma-flow.md and prisma-flow-diagram.png.

The 63 generation studies ran from 2023 to 2026 and covered, among other fields, radiology, orthopaedics, endodontics, neurocritical care, psychiatry and mental health, ophthalmology, otolaryngology, physical therapy, nephrology, dentistry, spine surgery, trauma care and general biomedical topics. One hundred and forty eight model arms were extracted. Models included the GPT family (the largest group), Gemini, Claude, DeepSeek, Grok, Bard, Copilot and several research systems in one study of AI research agents (Shi 2026). Verification sources were PubMed most often, then Crossref, Google Scholar and publisher records, named per arm in extraction.csv. Funding and conflicts were extracted per study; vendor involvement in the primary studies was uncommon in print, and several studies declared nothing either way, which the extraction records as not reported rather than as none.

Risk of bias in included studies

Adjudicated overall judgements across the 63 generation studies: high risk in 30, some concerns in 33, low in none. In the original 24 studies (after the duplicate merge), real adjudication moved Athaluri 2023 and Sawamura 2024 from high to some concerns against the first version's mechanical rule; the remaining first version studies kept their categories. Verification conduct drove most high judgements: studies that never said who verified the references, or whether verification was duplicated or blinded, could not score well on domains 5 and 6, and domain 6 carried the most no information judgements of any domain. Unversioned models with no generation date drove domain 2 high judgements in three early studies. Two single domain cases stand out: Chelli 2024 kept a prompt because it yielded the most results, and Bernstein 2026 defined hallucination to include real studies not indexed in PubMed, which can inflate a fabricated count. The full summary sits in rob-summary.md.

Synthesis of results

Completely fabricated. One hundred and thirteen arms from 50 studies carried a printed fabricated count, covering 22,461 generated references, of which 4,716 were fabricated (crude 21.0 percent). The random effects pooled proportion was 19.4 percent (95% CI 14.9 to 24.8), tau squared 2.59, I squared 97.1 percent, prediction interval 1.0 to 85.6 percent. The per study sensitivity pool gave 24.7 percent (18.2 to 32.5). Removing preprints gave 19.8 percent (15.1 to 25.6). Leave one family out pools sat between 15.9 and 21.1 percent, so no

single family carries the result. In the low or some concerns sensitivity pool (69 arms, 29 studies) the estimate was 19.1 percent (13.8 to 25.8), which says the headline figure is not an artefact of the weakest studies. The first version's estimate on the uncorrected dataset was 18.6 percent (12.8 to 26.3); the corrected estimate sits inside that interval with a narrower one of its own, on more than twice the arms.

Any major inaccuracy. Twenty arms from 14 studies, 2,668 references, 562 with a major inaccuracy (crude 21.1 percent). Pooled 16.5 percent (5.7 to 39.2), tau squared 6.48, I squared 96.7 percent, prediction interval 0.1 to 98.0 percent. The per study pool gave 26.8 percent (8.9 to 57.8). This pool changed the most in the robustness pass: settlement removed the first version's two largest contributors (Dhaimade 2026, whose major counts rested on a panel level category that is not a reference count, and Athaluri 2023, whose summed categories did not match the definition exactly) and added arms whose printed categories did map (among them Metze 2024 at 194 of 260). The first version's 36.0 percent should not be quoted; it rested on counts that do not survive a definition check. Even corrected, the pool is thin, because most studies graded errors on their own scales and only 14 printed a count that mapped exactly to the review definition, and the GRADE rating reflects that.

Fully accurate. Fifty eight arms from 28 studies, 14,224 references, 6,089 fully accurate (crude 42.8 percent). Pooled 37.0 percent (27.2 to 47.9), tau squared 2.89, I squared 98.2 percent, prediction interval 1.9 to 94.8 percent. The per study pool gave 38.2 percent (28.8 to 48.6), close to the arm level pool in the corrected set. Removing preprints gave 35.0 percent (24.6 to 47.1). In the low or some concerns pool (43 arms, 20 studies) the estimate was 39.1 percent (28.4 to 50.9). The first version's estimate was 29.4 percent (19.5 to 41.7); settlement blanked the loose accuracy counts of Civelekler 2026, Mugaanyi 2024 and Singh 2026, whose printed categories measured existence or relevance rather than accuracy on every checked field, and the enlarged study set pulled the estimate up.

Subgroups. By model family (family labels normalised across reports for this analysis), fabricated proportions pooled at 5.9 percent for Claude arms (9 arms, 1.8 to 17.7), 24.4 percent for GPT arms (59 arms, 16.2 to 35.1), 21.7 percent for Gemini arms (15 arms, 10.4 to 39.9) and 29.6 percent for DeepSeek arms (10 arms, 12.7 to 54.8). Fully accurate proportions by family were 65.9 percent for Claude (9 arms, 36.1 to 86.8), 33.4 percent for GPT (38 arms, 22.2 to 46.9) and 23.6 percent for Gemini (7 arms, 8.7 to 49.9). These family pools mix model generations and study topics, and the confidence intervals for Gemini and DeepSeek are wide; the family split is a description of this literature, not a league table of current products. By retrieval: arms with retrieval, browsing or database access active pooled 5.8 percent fabricated (26 arms, 12 studies) against 31.0 percent for arms with retrieval off (50

arms) and 19.9 percent where retrieval status was not reported (37 arms). Fully accurate with retrieval was 50.7 percent (12 arms) against 34.3 percent with retrieval off (27 arms), and major inaccuracy with retrieval was 1.2 percent (6 arms) against 37.7 percent off (11 arms). The retrieval gap is now carried by 26 arms rather than five, and it is still the least settled number in the review, because retrieval arms cluster in later studies with later models. By publication type, fabricated pooled 19.8 percent in peer reviewed reports (106 arms) and 13.5 percent in preprints (7 arms), a split that tracks model year more than editorial process and is not read as a quality signal. List length and model year subgroups are reported as not estimable in the Methods, with the arms that carried a printed requested count too few and too one sided (all above 10) to split.

Audit stratum, reported apart and never pooled. Yin 2026 audited about 111 million references in 2.5 million papers across arXiv, bioRxiv, SSRN and PubMed Central, and estimated about 146,932 non-existent references in 2025, with rates rising sharply after mid 2024. Within the search set, the Topaz 2026 Lancet audit was identified and used in the Background; it sits in the background list rather than the extracted audit stratum because this review's audit extraction was completed for one large scale audit. The stratum answers RQ7 in one direction: generation errors are not staying in the lab. What proportion of published fabrications came from LLMs, as against human error, cannot be read off either audit, and this review does not claim it.

Studies that could not enter a pool. Thirteen of the 63 studies contributed no arm to the fabricated pool, most often because they printed percentages without counts, graded accuracy on a composite score such as the Reference Hallucination Score without a whole reference fabricated count (Aljamaan 2024, Tas 2026), or reported accuracy in a form that did not map to the review definitions after settlement (among the new studies, Liu 2026 and Woodrow 2025; Metze 2024 entered the pools in the robustness pass once its percentages were settled as uniquely derivable counts). Their results are in extraction.csv and extraction-outcomes.csv and are described here so the pools are not mistaken for the whole included set.

Forest plots are printed after the references in this manuscript: one by-study plot per outcome (Figures 1 to 3), with the random-effects pooled proportion drawn as a diamond and the prediction interval as a capped line under it, and arm-level panels in the appendix figures. The plot files are forest-fabricated-bystudy.png, forest-fully_accurate-bystudy.png and forest-major_inaccuracy-bystudy.png for the by-study plots, and forest-fabricated-arms-1 to 6.png, forest-fully_accurate-arms-1 to 3.png and forest-major_inaccuracy-arms-1.png for the arm panels, all produced by forest_plots_v2.py. An earlier plot build (forest_plots.py,

files forest-fabricated.png, forest-major_inaccuracy.png and forest-fully_accurate.png) drew no summary diamond and its images overflowed the page in the first PDF, so the major inaccuracy plot was also left out of that PDF; the fault and the fix are logged in llm-team-log.md, and the first-build files are kept in the repository for the record.

Discussion

Summary of main results

Across 113 model arms and 22,461 checked references, 19.4 percent of LLM generated biomedical references were completely fabricated (95% CI 14.9 to 24.8). Across 58 arms and 14,224 references, 37.0 percent were fully accurate (27.2 to 47.9). A reference list from a model in these studies was therefore still more likely to contain an invented paper than to be clean throughout, and the middle ground, real papers with wrong details, held the largest share of all. The major inaccuracy pool, 16.5 percent across 14 studies, points the same way on thinner data.

The averages are the least useful part of those numbers, because the prediction interval for fabrication runs from 1.0 to 85.6 percent. Within a single study that held topic and verification constant and varied only the model, fabrication ran from 15 of 480 references to 291 of 480 (Gunes 2026). A purchasing decision, an editorial policy or a personal habit built on the pooled 19.4 percent would be built on a number that describes almost no actual setup. What the review can support is narrower and stronger: fabrication is a property of a setup, and it changes by an order of magnitude between setups. In the corrected set, exactly two of 148 arms were error free across a full list (OpenEvidence in McLaughlin 2026, 120 of 120 fully accurate; one 20 reference ChatGPT-4 arm in Zhu 2025), and both sat at the retrieval grounded end of the literature.

Retrieval is the one setup feature that separated cleanly. Arms with browsing, retrieval or database access pooled 5.8 percent fabricated (26 arms) against 31.0 percent with retrieval off (50 arms), and Pastucha 2026 shows the pattern inside one study, where turning web search on cut fabricated counts in the same model family from 82 of 396 and 103 of 530 to 7 of 296 and 4 of 301. Twenty six retrieval arms still cannot settle a claim of that size, and retrieval arms came from later studies with later models, so generation and retrieval effects are tangled. The claim the data carry is that the gap is large enough, and consistent enough across the studies that tested it directly, that any future experiment which fails to state its retrieval status is wasting its readers' time. Thirty seven of 113 fabricated arms did not report retrieval status at all.

The family split will be the part of this review that gets quoted, so it gets its caveat in the same paragraph as its numbers. Claude arms pooled 5.9 percent fabricated and 65.9 percent fully accurate; GPT arms 24.4 percent and 33.4 percent; Gemini arms 21.7 percent and 23.6 percent. The Claude arms are nine, they cluster in later studies (Adiguzel 2026 found 1 fabricated reference in 150 for Claude Sonnet 4.6 and 1 in 150 for ChatGPT 5, with the gap sitting in partial accuracy and DOI errors rather than invention), and family pools mix generations that differ more from each other than families do. Anyone quoting the family figures without the arm counts and intervals is misusing the review.

The audit stratum turns a laboratory finding into a publishing problem. Yin 2026 estimated about 146,932 non-existent references across four large corpora in 2025, and Topaz 2026 counted fabricated references in about 1 in 277 published biomedical papers in early 2026, up from about 1 in 2,828 in 2023. Set against a pooled experimental fabrication rate of 19.4 percent per generated reference, the print rates look small, and they should: most published lists are human written or human checked, and peer review catches some invented entries. The direction is the point. The reservoir of generated, unchecked lists is large and growing, and the audits show the spill has started.

Limitations of the evidence included in the review

No included study sat at low risk of bias on the adapted tool, and 30 of 63 sat at high risk. The dominant weakness was not fraud or sloppiness in generation; it was silence about verification. A reader often cannot tell who checked the references, against what, with what rule for a near match, or whether a second checker ever saw them. Where verification is the outcome measure, that silence caps what any synthesis can claim, and it capped this one at very low certainty on all three critical outcomes. Heterogeneity stayed above 95 percent in every main pool and no pre-specified subgroup explained it away, because studies differed in topic difficulty, prompt design, list length and grading stringency at the same time as they differed in model. Definition drift is real: one study counted a reference as hallucinated where any two of title, first author and year were wrong (Chelli 2024), another counted fabricated and misrepresented together (Singh 2026), and Bernstein 2026 counted real papers outside PubMed as hallucinated. Each definition is defensible in its own paper. Pooled, they measure a family of related failures rather than one failure, and the manuscript says so wherever the numbers appear.

Limitations of the review process

This is a single author review built with LLM assistance, and its process limits are stated plainly. Screening used two LLM rule sets, not two human screeners; agreement was high and kappa was low for prevalence reasons, and both figures are published. The first version

settled screening disagreements with a coded rule and risk of bias disagreements with a more-cautious-wins rule, and extracted once with a verification pass; the robustness pass replaced all three with adjudication and extraction by reading, and the corrected numbers in this version rest on those readings, but the underlying screeners, extractors and assessors are still LLM agents working apart under written rules, not human reviewers, and a human team might read borderline abstracts and definitions differently. The search could not include Embase, Scopus or Web of Science, could not retrieve arXiv or Semantic Scholar past their rate limits on either attempt, and could not complete full OpenAlex retrieval for the same reason; it took in a very broad Europe PMC set whose PMCID field proved unreliable for a number of records, a fault that cost four reports their full text assessment in the first pass and recurred in the pass, and is reported in the team log rather than buried. One hundred and fifteen reports that passed screening were never retrieved, and some of them are likely eligible; the included set is therefore the open access visible part of this literature, which may tilt it toward better reported studies. Extraction rested on printed counts; where a paper printed only percentages, its arms stayed out of the pools unless exactly one whole count reproduced the printed percentage, which favours studies that reported completely.

Agreements and disagreements with other studies or reviews

No earlier synthesis pooled these experiments, so there is no direct comparator for the headline figures. The findings agree with the mitigation review of Basu and Huynh (44 studies), which found retrieval approaches cutting hallucinations by 30 to 50 percent in the studies it covered; this review's retrieval split, 5.8 against 31.0 percent fabricated, is larger than that range, so the agreement is in direction, not size. The findings sit uneasily with vendor facing claims that citation accuracy in current models is a solved problem: the best single arm in this set (Claude Sonnet 4.6 in Adiguzel 2026, 1 fabricated in 150) coexists with a 2026 arm at 98 fabricated in 149 (UcarGundogar 2026, DeepSeek R1), both in peer reviewed reports from the same year. The review also disagrees, in method, with any reading of the audits that treats print prevalence as a model property: the audits measure a pipeline that includes authors, co-authors, reviewers and editors, and the experiments measure the model step of that pipeline.

Authors' conclusions

Implications for practice

This review makes no clinical recommendation and nothing in it bears on the care of any patient. For the practice of writing, reviewing and editing research, the conclusions are

these. A generated reference list should be treated as unverified text until every entry has been checked against a database, because 19.4 percent of generated references in these experiments did not exist and only 37.0 percent were fully accurate. Journals and institutions that ask authors to declare AI use could ask a sharper question at the same time: how was each AI assisted reference verified, and by whom. Librarians and research support staff are the group best placed to run that check at scale, which is why the journal route for this review targets their literature.

Implications for research

Four things would lift the next version of this evidence base out of very low certainty. First, report verification like a method, not an afterthought: who checked, against which sources, blind to model or not, with a stated rule for partial matches and a second checker on at least a sample. Second, print counts, not only percentages, per model arm, with the denominator, the prompt, the model version, the generation date and the retrieval status on the same page. Third, run retrieval as an experimental factor inside one study on shared topics, as Pastucha 2026 began to do, rather than leaving it as a between study comparison tangled with model year. Fourth, register these experiments and report every model and topic run, so the published record stops being a selection of the runs that finished. A repeat of this review in twelve months, on a literature that did those four things, could support a moderate certainty rating. On the current literature, it cannot.

Declarations

Funding: none. This review was carried out without funding from any source.

Conflicts of interest: none. The author has no industry affiliations of any kind and is fully independent.

Author contributions: Raphael Malikian set the question and the eligibility rules, settled flagged items, checked the numbers that entered the pools against their sources, and wrote and takes responsibility for the final text. Screening, extraction, risk of bias assessment, analysis and drafting support were carried out by an LLM team working under his direction, as described in llm-team-log.md.

AI assistance note: AI tools were used for literature searching, screening, data extraction, analysis and drafting support in the preparation of this review, under the direction of the author, who reviewed the full text and takes full responsibility for it. The use of AI inside the review process is reported in full in llm-team-log.md, including agreement statistics, disagreement counts and adjudication rules.

Data, code and other materials: the protocol, search log and results, screening files, extraction files (including both independent passes and the settlement log from the robustness pass), risk of bias assessments, analysis datasets (analysis-fabricated.csv, analysis-major_inaccuracy.csv, analysis-fully_accurate.csv), analysis code (analysis.py, with analysis_norm.py carrying the controlled vocabulary normalisation used for subgroup analyses in the corrected version, and forest_plots_v2.py) and forest plots are published with this review in the GitHub repository and the Zenodo package. Licence: CC BY 4.0 for text and data, MIT for code.

Registration and protocol: protocol.md in this series, version 1.0, 22 October 2026. No PROSPERO number was issued before analysis; a PROSPERO-ready form (prospero-form.md) and an OSF record are the registration route, and neither is backdated.

References

Included generation studies (63): Adıgüzel 2026 (dentistry, endodontics); Aiumtrakul 2023 (nephrology); Aljamaan 2024 (mixed biomedical); Athaluri 2023 (mixed biomedical); Awosanya 2024 (orthopaedic surgery); Bernstein 2026 (orthopaedic surgery, AAOS guidelines); Blasingame 2025 (mixed biomedical); Bridges 2025 (mixed biomedical); Chelli 2024 (systematic review references); Choi 2023 (pharmacology); Citir 2025 (dentistry); Civelekler 2026 (ophthalmology); Dhaimade 2026 (examination questions); DiFebo 2026 (cardiology); Eminovic 2026 (radiology and imaging); Ettien 2026 (mixed biomedical); Ghanem 2024 (orthopaedic trauma); Gravel 2023 (mixed biomedical); Gwon 2024 (urology); Güneş 2026 (radiology); Hack 2026 (otolaryngology); Hasnain 2025 (ophthalmology); Hueber 2023 (rheumatology); Idan 2025 (paediatrics and related fields); Jedrzejczak 2023 (audiology); Kang 2026 (paediatrics); Kaygisiz 2025 (dentistry); Knee 2025 (orthopaedic surgery); Knee 2026 (orthopaedic surgery); Kutluay 2026 (orthopaedics and traumatology); Lin 2025 (nephrology); Linardon 2025 (psychiatry and mental health); Liu 2026 (dentistry); McCavitt 2026 (orthopaedic surgery); McLaughlin 2026 (spine surgery); Metze 2024 (bibliographic research by professors); Mikus 2024 (pharmacology); Moskatel 2023 (neurology); Mugaanyi 2024 (cross disciplinary, biomedical subset); Naldi 2024 (dermatology); Nieves-Lopez 2026 (foot and ankle surgery); Ohlsen 2026 (orthopaedic surgery); Ok 2026 (urology); Pan 2026 (nursing and allied health); Pastucha 2026 (otorhinolaryngology; journal version of the study first reported as Jedrzejczak 2024, merged at study level in the robustness pass); Rao 2026 (plastic surgery); Safrai 2024 (reproductive medicine); Sawamura 2024 (physical therapy); Schumacher 2024 (ophthalmology); Schwartzman 2025 (orthopaedic surgery); Seifi 2026 (neurocritical care); Shi 2026 (AI research systems); Sina 2024 (otolaryngology); Singh 2026 (orthopaedic

surgery); Suppadungsuk 2023 (nephrology); Tas 2026 (reference hallucination framework); Teke 2026 (dentistry); Uçar 2026 (dermatology); UçarGündoğar 2026 (dentistry); Warren 2024 (urology); Woodrow 2025 (orthopaedic surgery); Zhou 2026 (ophthalmology); Zhu 2025 (public health and epidemiology). Full citations, DOIs and record identifiers for every included study sit in extraction.csv and search-results.csv, which form part of this review's published data.

Audit stratum: Yin 2026, LLM hallucinations in the wild: large scale evidence from non-existent citations (analysis of about 111 million references in 2.5 million papers across arXiv, bioRxiv, SSRN and PubMed Central).

Other references: Topaz M et al. Fabricated citations: an audit across 2.5 million biomedical papers. The Lancet. 2026;407:1779-81. Zhao et al. 2025 (ambient scribe meta-analysis, series background). Basu S, Huynh B. Mitigating hallucinations in healthcare AI: a systematic review of evidence-based strategies. BMC Health Services Research. 2026. Cochrane Handbook for Systematic Reviews of Interventions, version 6.5, 2024. Page MJ et al. PRISMA 2020 statement. BMJ. 2021;372:n71. RAISE position statement on responsible AI use in evidence synthesis, Cochrane, Campbell, JBI and CEE, 2025.

Appendices

Appendix A: full search strings as run, in search-log.md. Appendix B: ready to run strings for sources pending access, in search-appendix-pending.md. Appendix C: characteristics of included studies, in extraction.csv and extraction-outcomes.csv. Appendix D: characteristics of excluded full text reports, with one reason each, in fulltext-screening.csv. Appendix E: PRISMA 2020 checklist, in journal-package/prisma-2020-checklist.md.

Forest plot: completely fabricated, by study

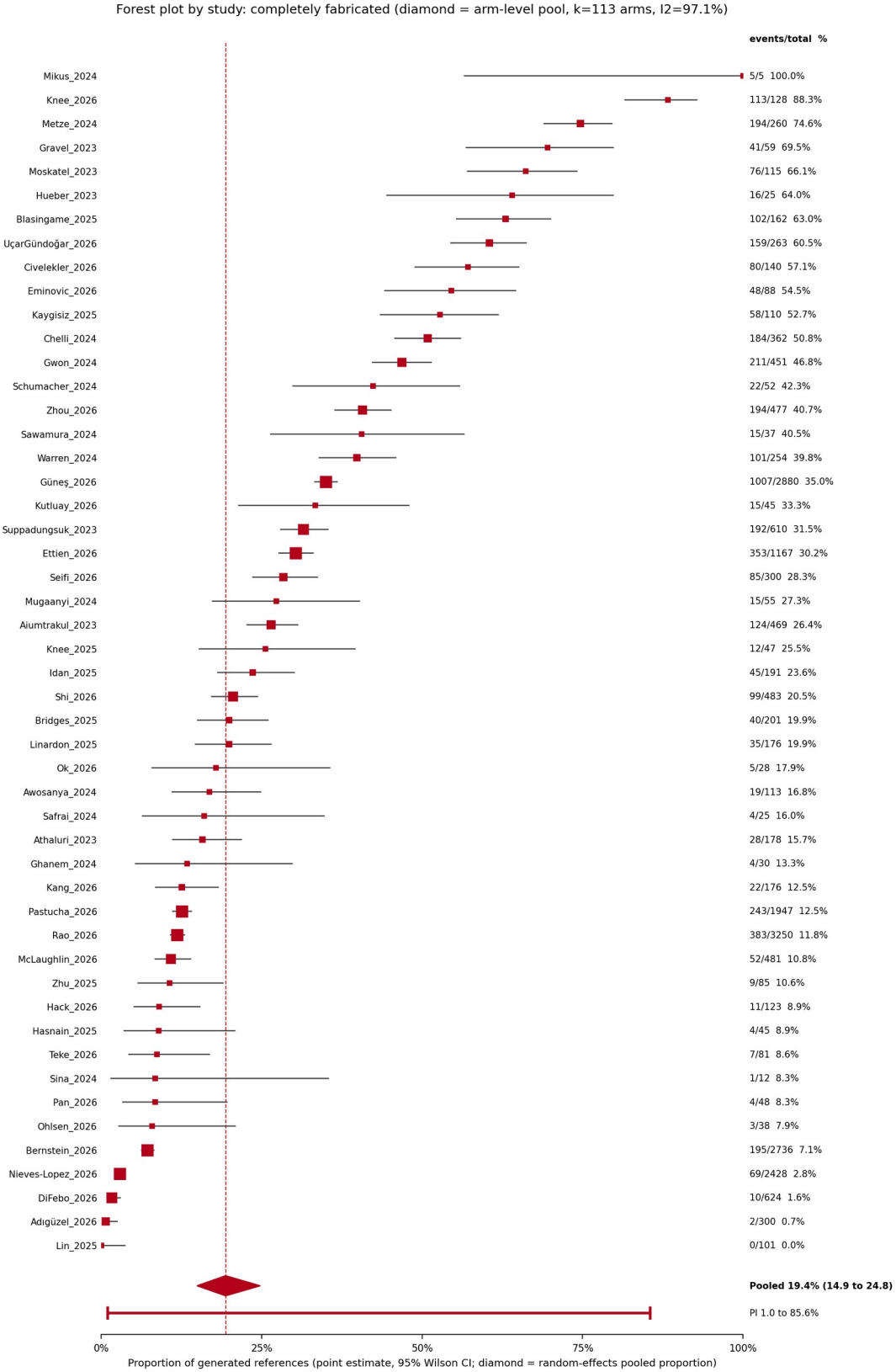


Figure 1. One row per study (study totals across its model arms). The diamond is the arm-level random-effects pooled proportion, 19.4% (95% CI 14.9 to 24.8); the capped line under it is the 95% prediction interval, 1.0 to 85.6%. The per-study sensitivity pool was 24.7%. Corrected version, robustness pass 23 October 2026 (first version: 18.6%, 51 arms).

Forest plot: fully accurate, by study

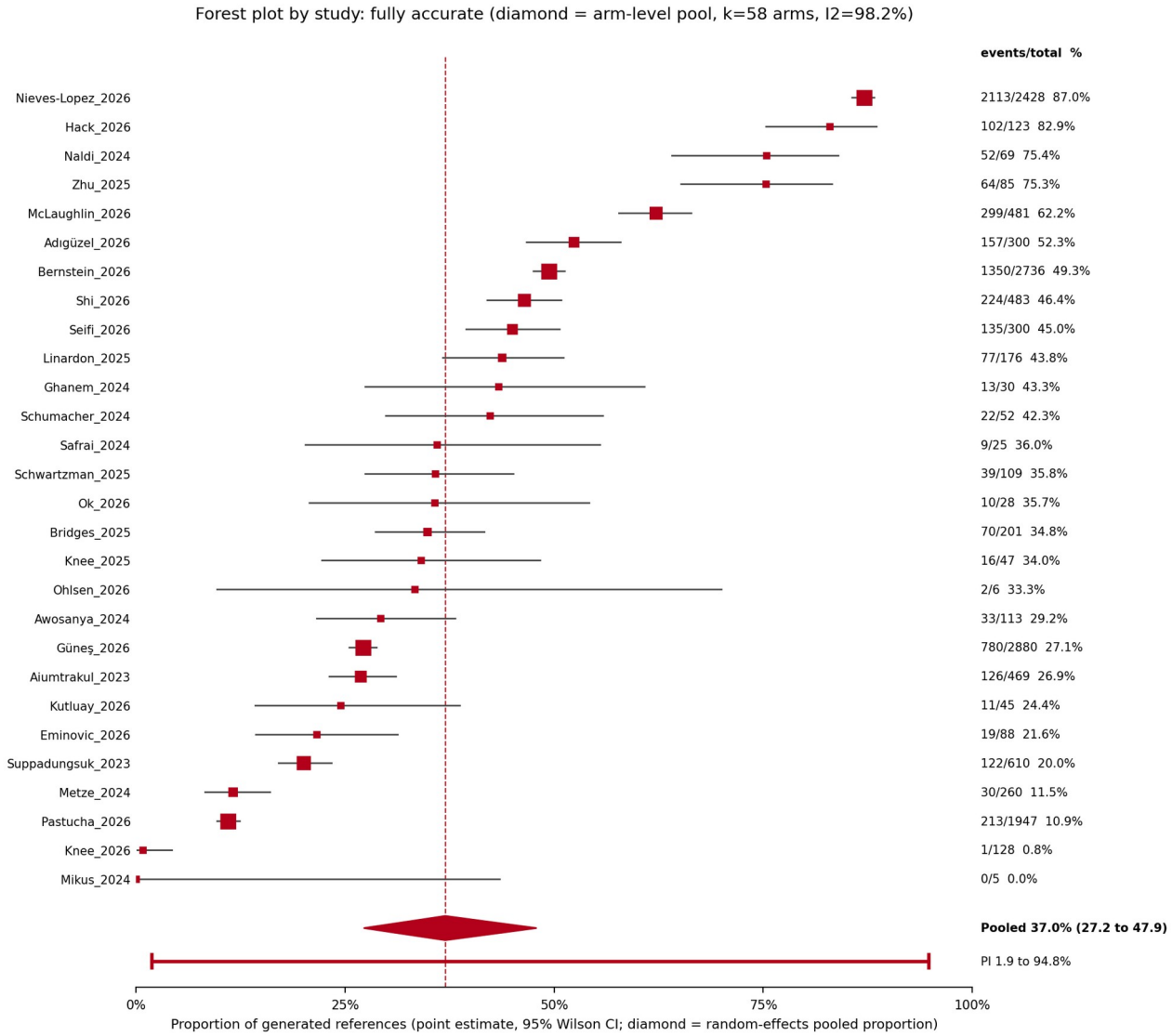


Figure 2. One row per study. Diamond: arm-level pooled proportion fully accurate, 37.0% (95% CI 27.2 to 47.9); capped line: 95% prediction interval, 1.9 to 94.8%. Per-study sensitivity pool 38.2%.

Forest plot: any major inaccuracy, by study

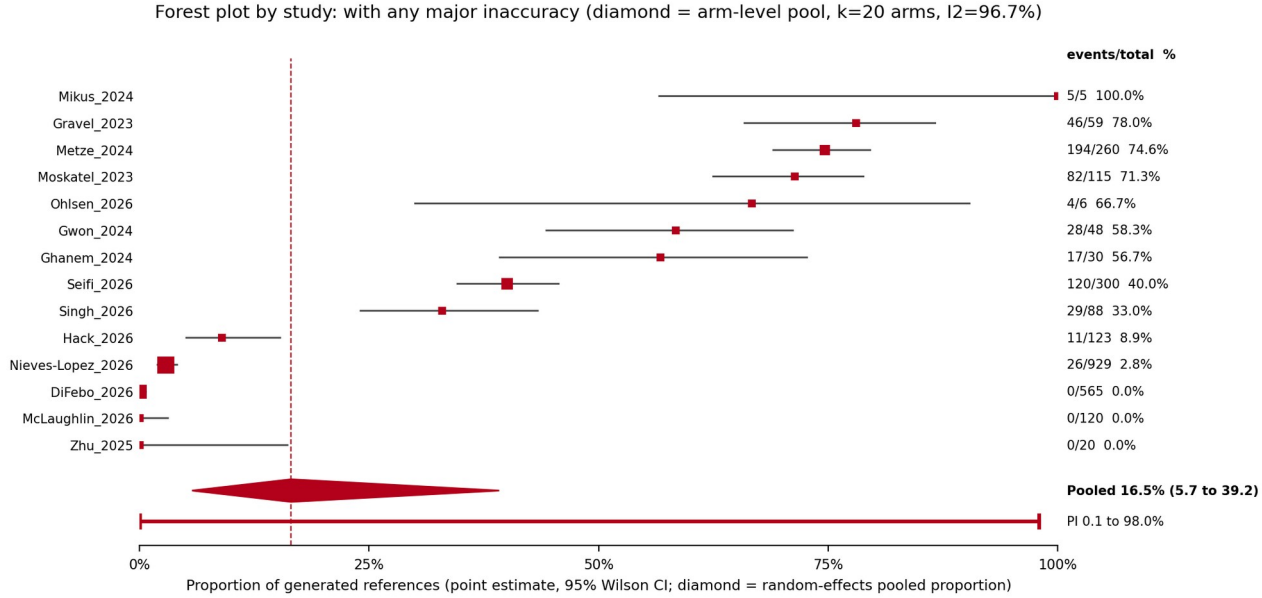
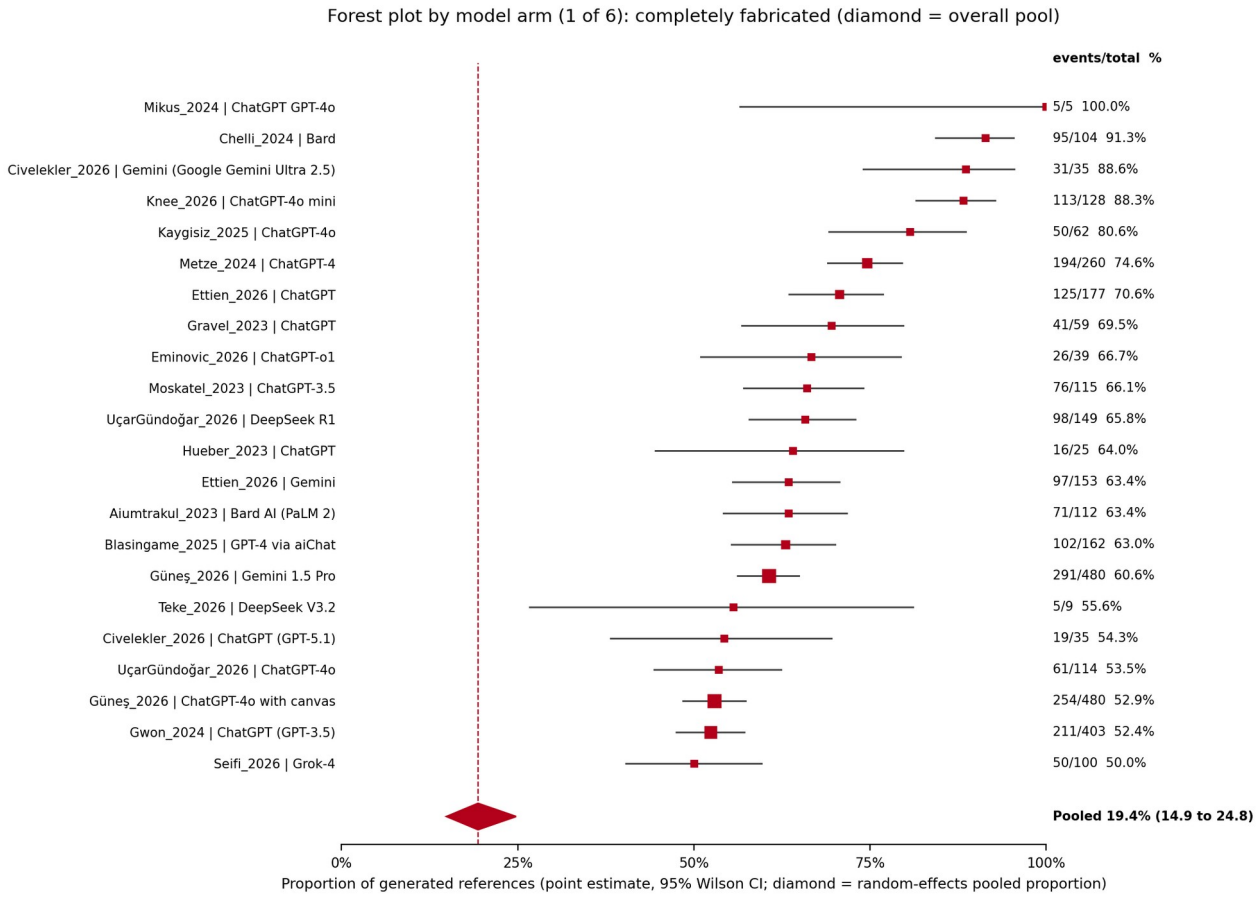


Figure 3. One row per study. Diamond: pooled proportion with any major inaccuracy including fabricated, 16.5% (95% CI 5.7 to 39.2), from 20 arms in 14 studies. The first version's figure for this outcome (36.0%, 9 arms in 5 studies) rested on counts that settlement found did not map to the definition and is superseded.

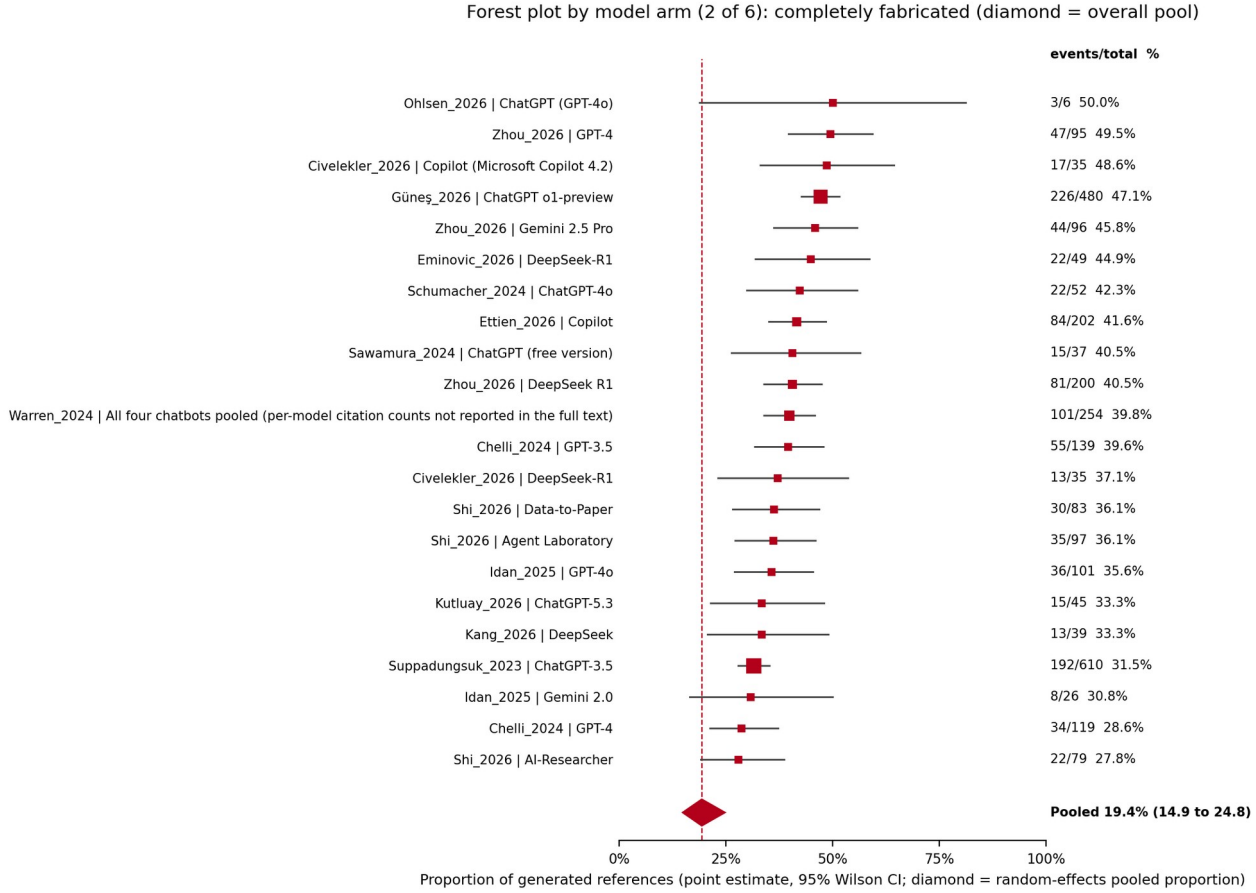
Appendix figures: forest plots by model arm

Fabricated, arms 1 of 6



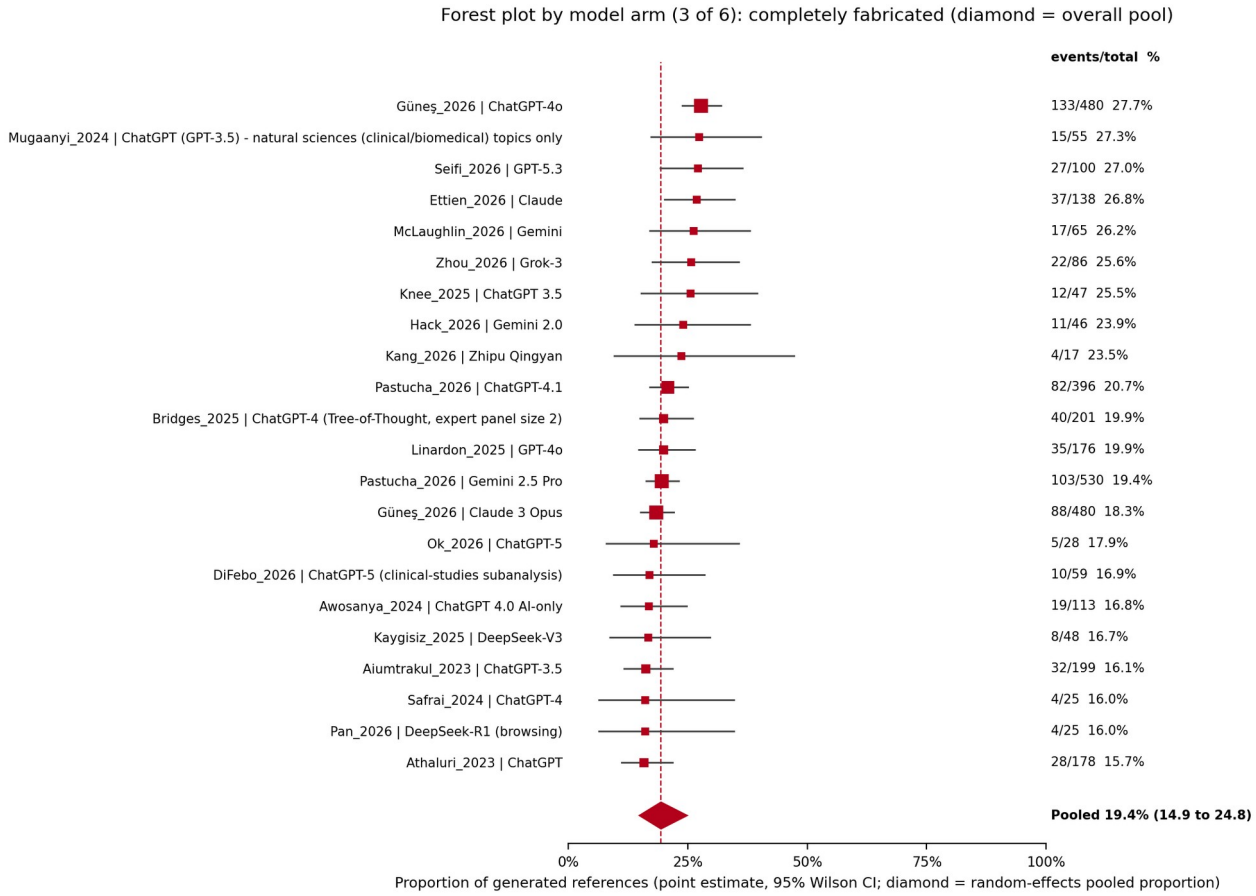
Fabricated, arms 1 of 6. The diamond on each panel is the overall arm-level pool.

Fabricated, arms 2 of 6



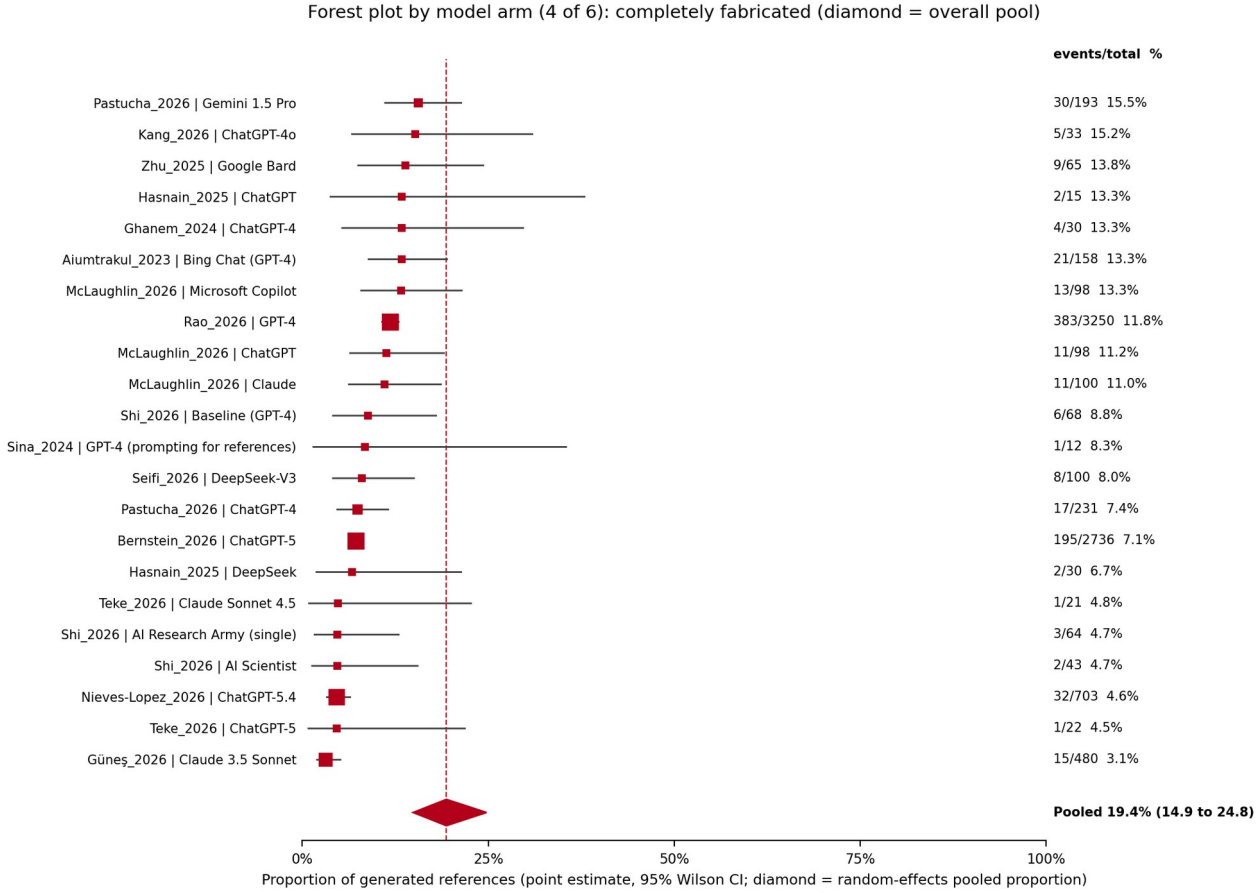
Fabricated, arms 2 of 6.

Fabricated, arms 3 of 6



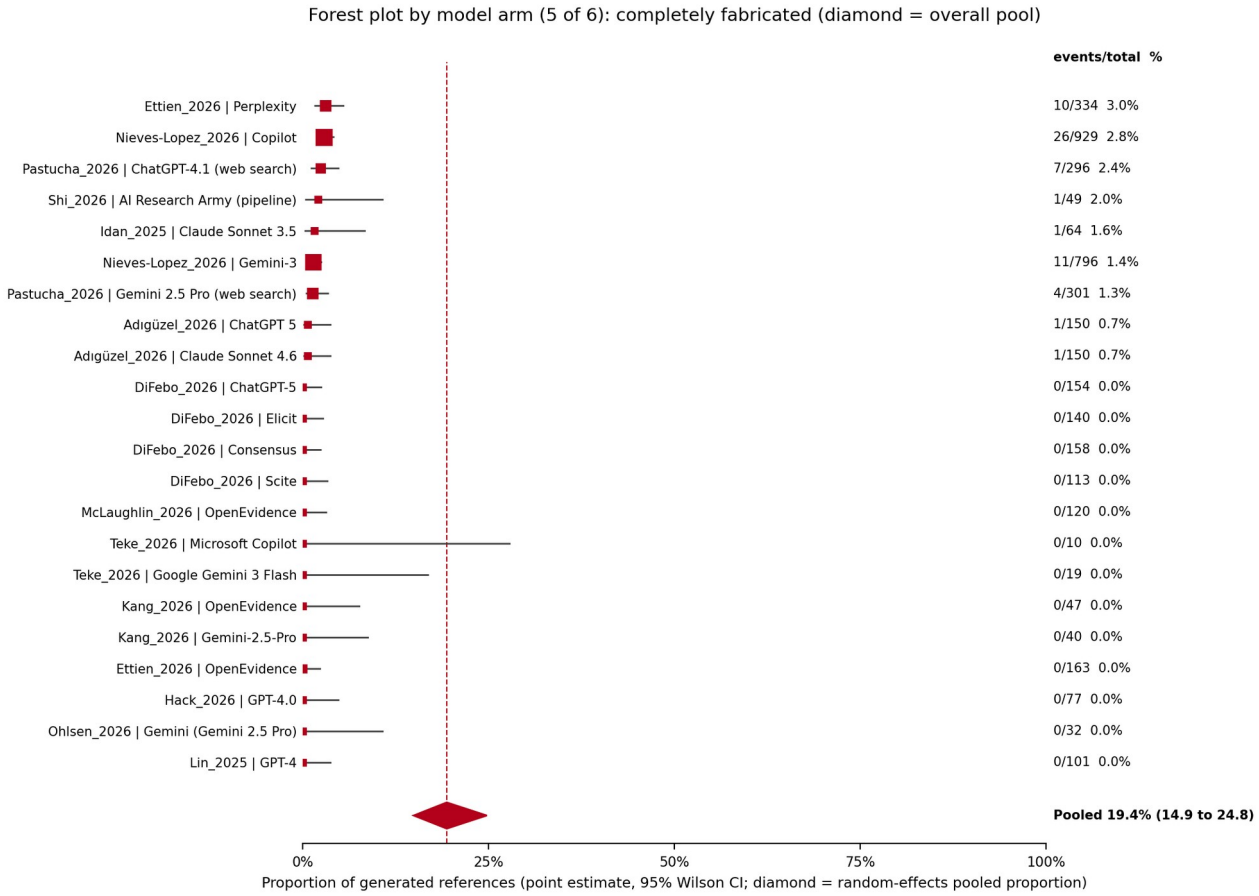
Fabricated, arms 3 of 6.

Fabricated, arms 4 of 6



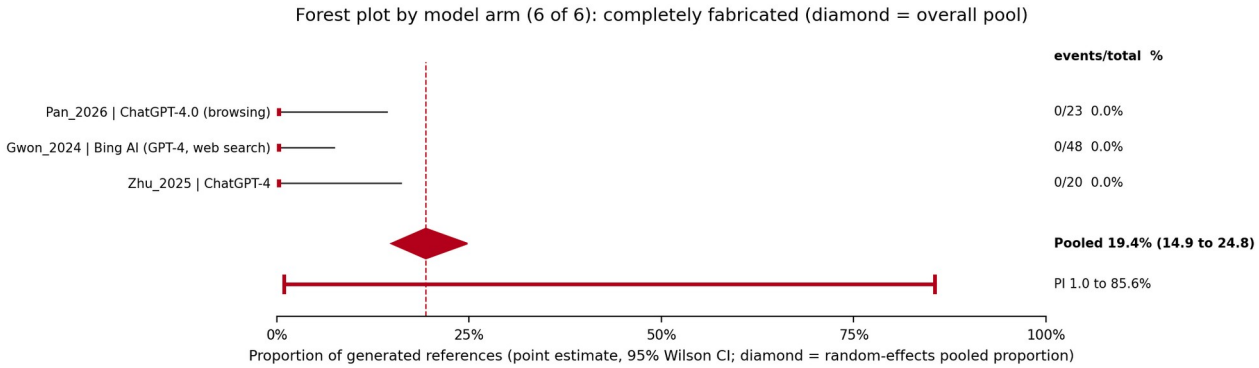
Fabricated, arms 4 of 6.

Fabricated, arms 5 of 6



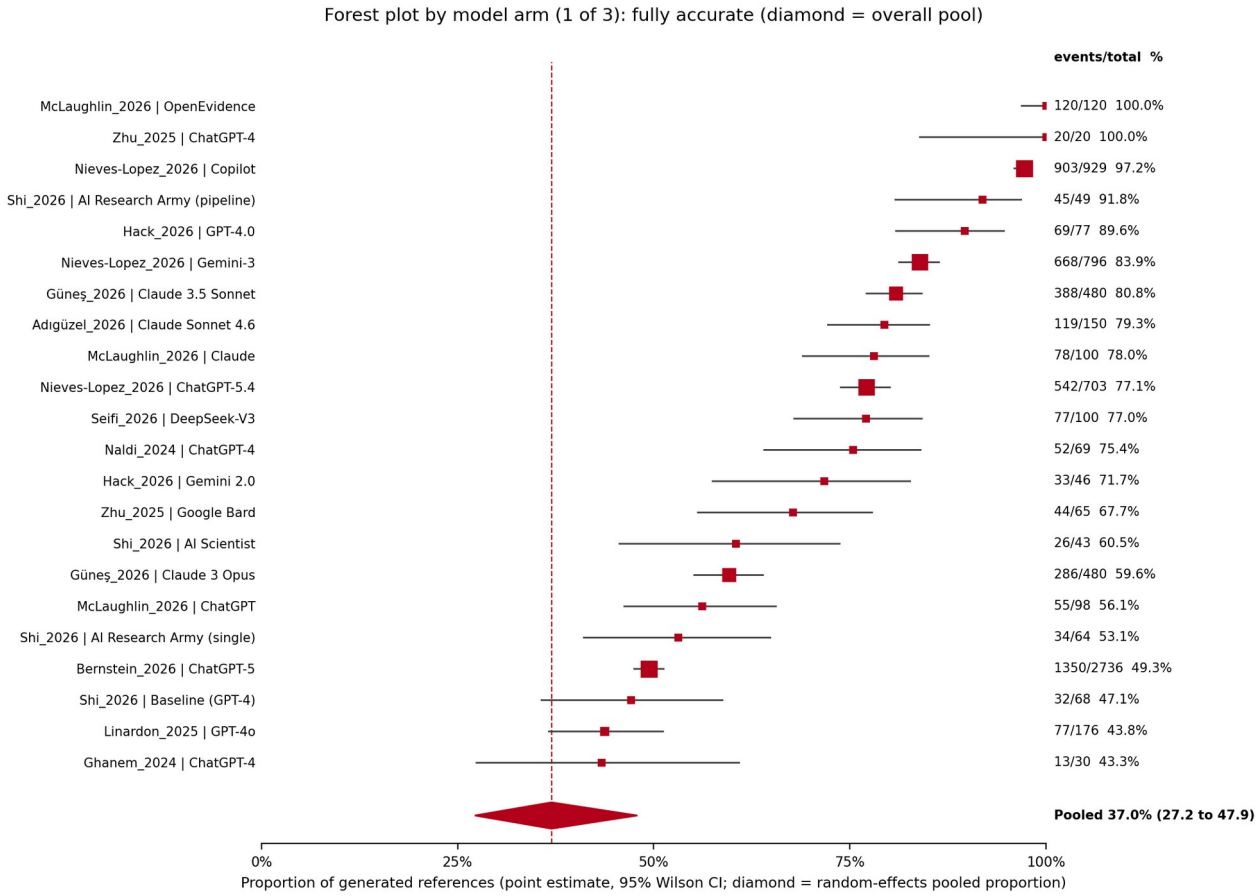
Fabricated, arms 5 of 6.

Fabricated, arms 6 of 6, with the prediction interval



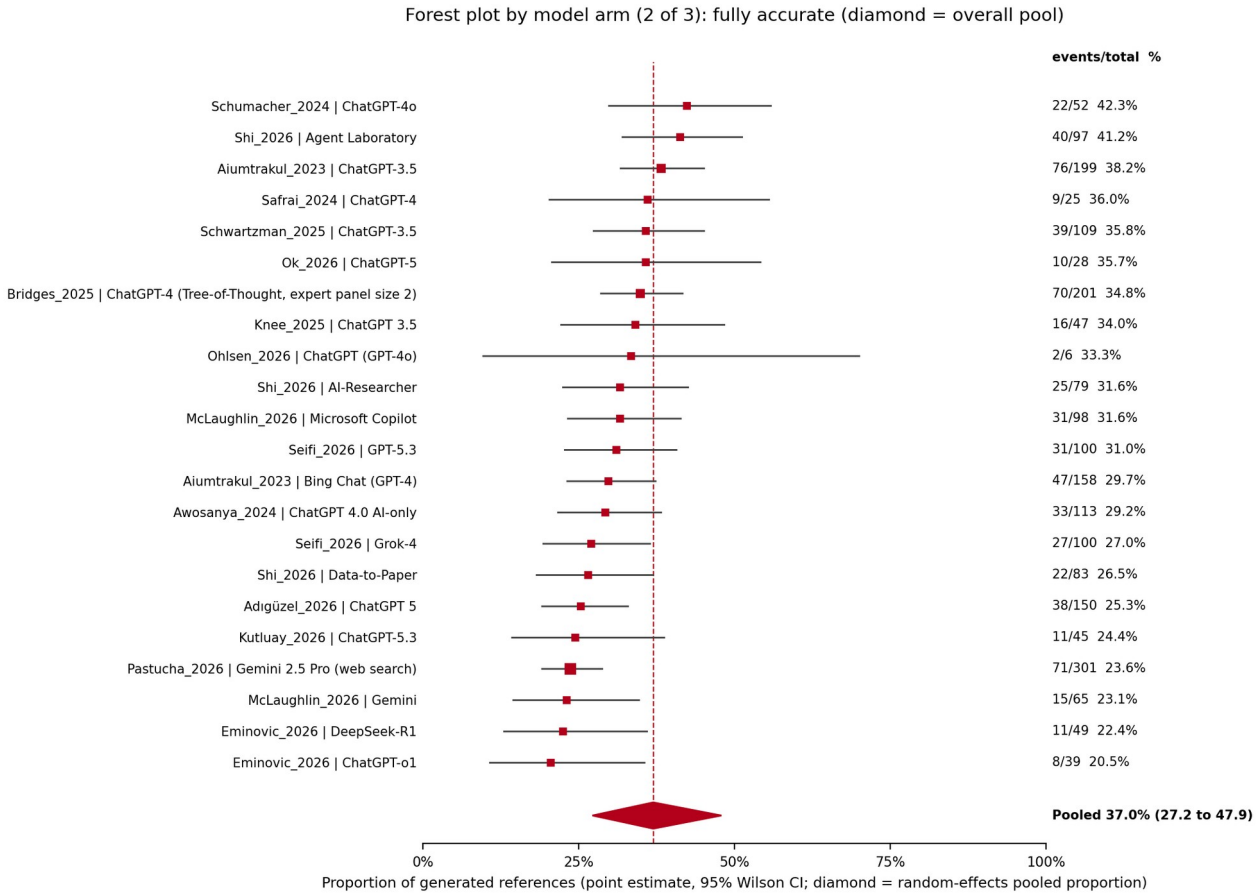
Fabricated, arms 6 of 6, with the prediction interval.

Fully accurate, arms 1 of 3



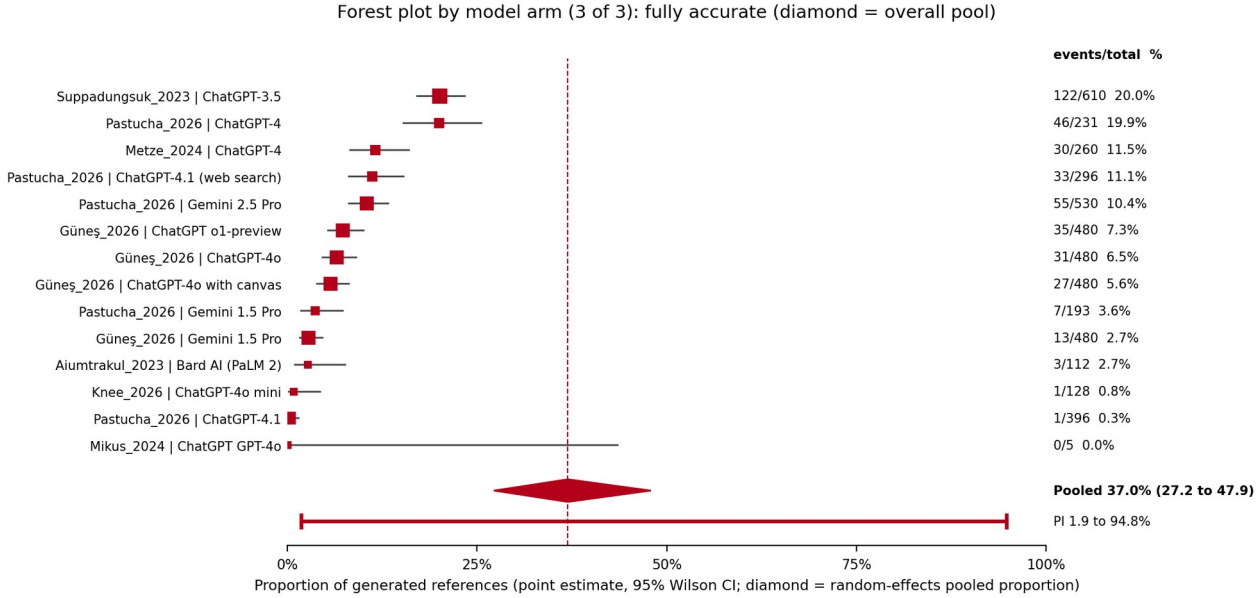
Fully accurate, arms 1 of 3.

Fully accurate, arms 2 of 3



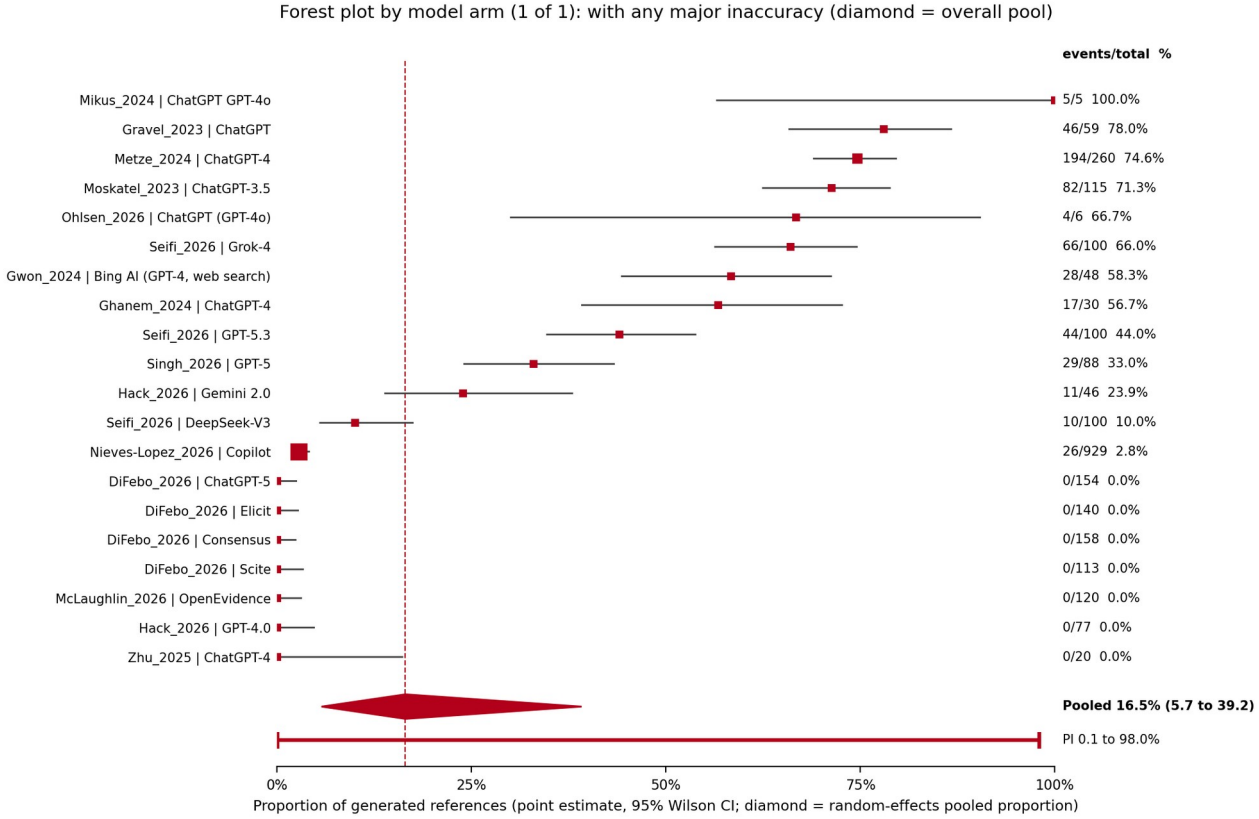
Fully accurate, arms 2 of 3.

Fully accurate, arms 3 of 3, with the prediction interval



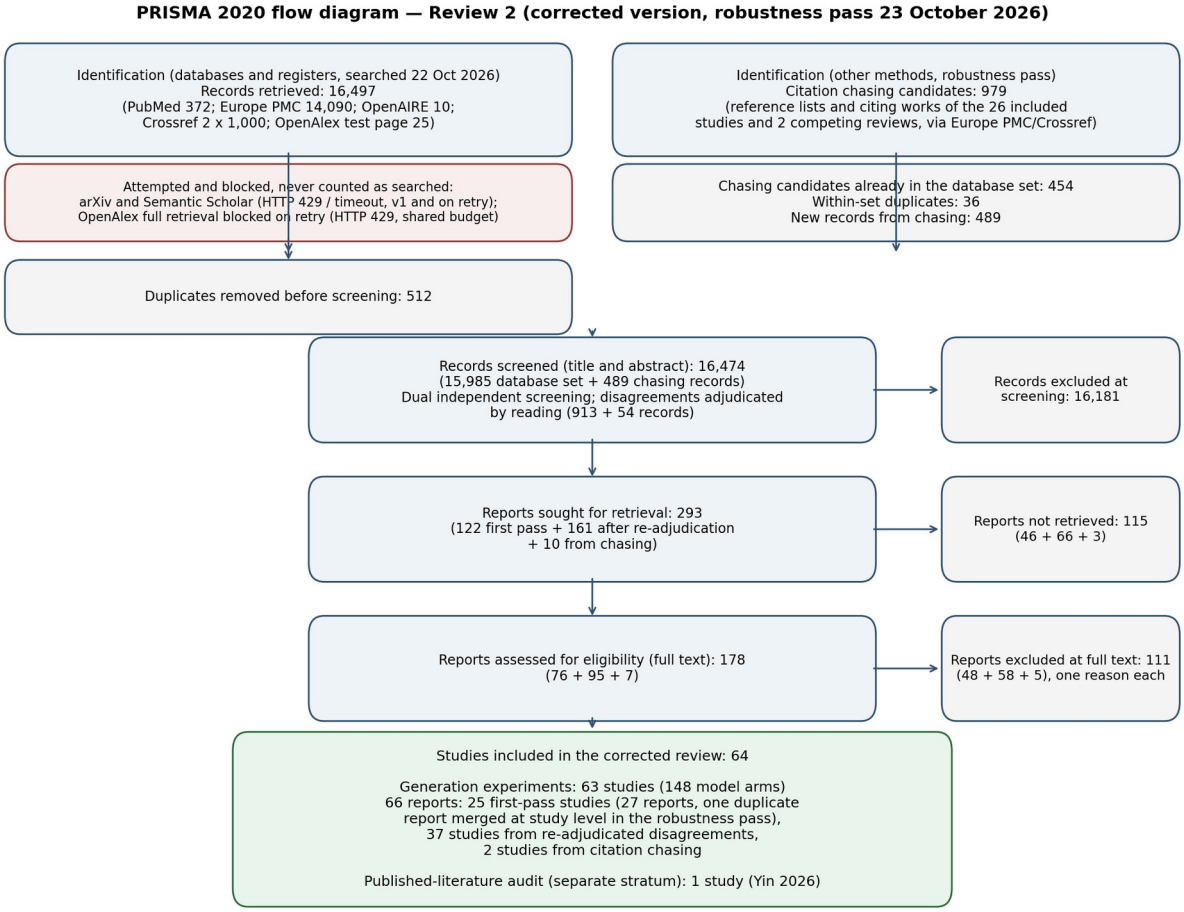
Fully accurate, arms 3 of 3, with the prediction interval.

Any major inaccuracy, all 20 arms



Any major inaccuracy, all 20 arms.

PRISMA flow diagram



First version (22 Oct 2026) for comparison: 15,985 screened; 122 sought; 76 assessed; 26 studies included (25 generation + 1 audit).